S T A N D E X I N T E R N A T I O N A L

Close Binder Tracker

Product Specifications & Requirements Document

Quarterly Close Binder File Submission Tracker

Version 2.0 | March 2026

C O N F I D E N T I A L

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1. Introduction

1.1 Purpose

This document defines the complete product specifications and functional requirements for the Standex Close Binder Tracker application. It serves as the authoritative reference for all features, workflows, data models, user roles, and non-functional requirements governing the system.

1.2 Background

Standex International operates approximately 77 business units across 6 segments worldwide. Each quarter, every business unit must submit up to 31 close binder files (trial balances, reconciliations, aging reports, revenue calculations, and compliance certifications) to Corporate Finance for review and approval.

Prior to this application, Corporate Finance had no centralized way to track whether a business unit had completed its submission, which files were missing, and whether missing files were genuinely not applicable or simply overlooked. The Close Binder Tracker solves this by providing a centralized portal with explicit acknowledgment of non-applicable items, creating an auditable record of every submission.

1.3 Scope

Dimension	Detail
Business Units	77 divisions across Electronics, Engraving, Engineering Technologies, Scientific, Specialty Solutions, and Corporate Services
Required Files	31 items per quarter (items 1–28, including 13a, 13b, 13c), varying by BU applicability
Fiscal Year Structure	4 quarters: Q1 ends Sep 30, Q2 ends Dec 31, Q3 ends Mar 31, Q4 ends Jun 30
User Roles	BU Submitter, BU Viewer, Corp Finance Reviewer, Corp Finance Approver, IT Admin
Hosting Platform	Microsoft Azure (Container Apps, PostgreSQL Flexible Server, Container Registry)
Monthly Cost	~$51/month (Container App $30, PostgreSQL $15, ACR $5, Storage $1)
Phase 2 Integrations	Azure AD SSO, SharePoint file sync, Azure Blob Storage for uploads

1.4 Definitions & Abbreviations

Term	Definition
BU	Business Unit — a Standex division or subsidiary identified by a 5-digit divisional code
Close Binder	The collection of quarterly financial files that each BU must submit to Corporate Finance
N/A	Not Applicable — a BU declares that a particular required file does not apply to them for a given quarter
Corp Finance	Standex Corporate Finance team responsible for reviewing and approving BU submissions

JWT	JSON Web Token — used for authentication in the production application
SAS URL	Shared Access Signature URL — time-limited Azure Blob Storage URL for file uploads (Phase 2)
Submission	The complete package of files (uploaded or marked N/A) from one BU for one quarter

2. User Roles & Permissions

The system enforces role-based access control (RBAC) with five distinct roles. Each role has specific permissions, and BU-scoped access is enforced at the API layer through the user_business_units junction table.

2.1 Role Definitions

Role	Description	Key Permissions
BU Submitter	Business unit finance personnel responsible for uploading and submitting close binder files	Upload files, mark N/A, submit to Corp Finance, withdraw pre-review submissions, view own BU data only
BU Viewer	Read-only access for BU personnel who need visibility but do not submit	View submission status and files for assigned BUs only; cannot upload, submit, or modify
Corp Reviewer	Corporate Finance team members who review submissions	View all BUs, review submitted files, reject submissions with reasons; cannot approve
Corp Approver	Senior Corporate Finance personnel authorized to approve	All Corp Reviewer permissions plus approve submissions, manage quarters and users
IT Admin	System administrator for user and configuration management	All Corp Approver permissions plus full admin access: create users, assign BUs, manage all settings

2.2 BU Access Control Model

Users are assigned to specific business units via the user_business_units junction table. A BU Submitter assigned to 5 BUs sees only those 5 BUs and their submissions. Corp Finance and IT Admin roles bypass the junction table and see all BUs.

Demo user assignments (production seed data):

User	Role	BU Count	Business Units
Jane Doe	Submitter	5	Electronics US, Mexico, ATC-Frost, Northlake, Horizon Scientific
Carlos Rivera	Submitter	7	Tenibac, SE Clinton, GS Eng, Mold-Tech SE/Midwest/Mexico, WCS Mexico
Akiko Tanaka	Submitter	8	Electronics Tianjin, Trading China, Kofu, Sanyu JP/SEPL, India, Meder Asia/Shanghai
Rainer Hombeck	Submitter	10	Engraving EMEA: France, Spain, Italy, Portugal, Germany, UK, Czech, Turkey, Romania, Solare
John Barker	Submitter	6	Engineering Tech: Spincraft WI/MA/UK, McStarlite, Custom Hoists/China
Sarah Chen	Viewer	5	Same as Jane Doe (view only — no upload/submit)
Mark Reynolds	Approver	All 36	Sees all BUs with submissions; can approve
Danielle Rangel	Reviewer	All 36	Sees all BUs; can review and reject but NOT approve
Lisa Wong	Reviewer	All 36	Same as Danielle
IT Admin	Admin	All 36	Full system access including Administration section

3. Required Close Binder Files

Each business unit must account for all 31 required files every quarter. For each file, the BU must either upload the document or explicitly acknowledge it as not applicable (N/A) for their business unit.

3.1 Complete File List

#	Category	Required File Name
1	General	Trial Balance for the Entire Quarter (If system allows, please provide Excel versions ONLY)
2	Cash	Cash - Bank Statement & Reconciliation
3	Receivables	A/R Complete Aging By Invoice & Detailed Reconciliation to GL
4	Receivables	A/R Bad Debt Reserve including calculation used to determine % allocation
5	Receivables	A/R Return Reserve & Lag Analysis
6	Inventory	Inventory Complete Aging and Detailed Reconciliation to GL
7	Inventory	Inventory E & O, Include Calculation Detail
8	Inventory	Inventory PPV, Include Calculation Detail
9	Inventory	Inventory Non-PPV, Include Calculation Detail
10	Inventory	Inventory LOCM, Include Calculation Detail
11	Payables	AP Complete Aging & Recon to GL
12	Accruals	Warranty Reserve & Lag Analysis and Detailed Calculation
13	Accruals	Other Accrued Expenses- (non-payroll/benefits)
13 a)	Accruals	Accrued Rebates, Including Detailed Calculation
13 b)	Accruals	Accrued Commissions, Including Detailed Calculation
13 c)	Accruals	Accrued Vacation, Including Detailed Calculation
14	Revenue	Sales Journal Reconciled to Gross Revenue in OS
15	General	Vender Master File Review
16	General	Customer Master File Review
17	General	Journal Entries - reviewed query of manual adjusting journal entries each month of the quarter
18	General	General Ledger Transaction Detail for the Quarter RECONCILED to the trial balance (provide supporting reconciliation documentation, if system allows, please provide Excel versions ONLY)
19	Revenue	Revenue - Percentage of Completion Calculation (include evidence of review and discussion)
20	Revenue	Revenue Cutoff - Reviewed Support, including shipping documents for Last 5 First 5 at Quarter End and additional selections as required by form
21	Revenue	Revenue Cutoff - Reviewed documentation to ensure completeness of Last 5 First 5.
22	Revenue	ASC 606 Overtime PoC Impact Calculation - YTD Metric
23	Revenue	ASC 606 General Impact Calculation - YTD Metric
24	Inventory	Cycle Count Reporting and Monitoring worksheet
25	Inventory	Cycle Count Last Count Date Report for Quarter
26	Inventory	Cycle Count Certification Form A.3.e

27	General	Lease Reconciliation Balance Sheet Template
28	Receivables	Allowance for Credit Losses Analysis - Quarterly

3.2 File Categories

Category	Items	Description
General	1, 15–18, 27	Trial balance, vendor/customer master files, journal entries, GL detail, lease reconciliation
Cash	2	Bank statements and reconciliations
Receivables	3–5, 28	A/R aging, bad debt reserves, return reserves, credit loss allowance
Inventory	6–10, 24–26	Inventory aging, E&O, PPV, non-PPV, LOCM, cycle count reports and certification
Payables	11	A/P aging and reconciliation to general ledger
Accruals	12–13c	Warranty reserves, accrued expenses, rebates, commissions, vacation
Revenue	14, 19–23	Sales journal, revenue percentage-of-completion, cutoff analysis, ASC 606 calculations

3.3 Due Date Schedule

File due dates are calculated relative to each quarter end date. The standard due date is D+11 business days (~15 calendar days) from quarter end. Certain items have accelerated due dates of D+7 business days (~10 calendar days) for Q4/year-end (items 1, 2, 3, 11, 18).

Quarter	End Date	Standard Due	Accelerated Due
FY2026 Q1	Sep 30, 2025	Oct 15, 2025	N/A
FY2026 Q2	Dec 31, 2025	Jan 15, 2026	N/A
FY2026 Q3	Mar 31, 2026	Apr 15, 2026	N/A
FY2026 Q4	Jun 30, 2026	Jul 15, 2026	Jul 10, 2026

4. Functional Requirements

4.1 Authentication & Authorization

FR-AUTH: User Authentication

- Production uses JWT token authentication with bcrypt-hashed passwords
- Login endpoint: POST /api/auth/login accepts email + password, returns JWT token (8-hour expiry)
- All API endpoints require a valid Bearer token in the Authorization header
- Rate limiting: 20 login attempts per 15 minutes per IP; 500 API requests per 15 minutes
- Forgot password flow: generates a single-use token (1-hour expiry), sends reset email via SMTP
- Password change endpoint for authenticated users
- Phase 2: Azure AD (Entra ID) SSO integration with MSAL.js frontend and passport-azure-ad backend

FR-RBAC: Role-Based Access Control

- API middleware extracts user role and BU assignments from JWT claims
- BU Submitters and Viewers see only their assigned BUs (enforced via SQL JOINs against user_business_units)
- Corp Finance roles (Reviewer, Approver) and IT Admin bypass BU restrictions and see all submissions
- Approve action restricted to corp_approver role only
- Reject action available to both corp_approver and corp_reviewer roles
- Admin endpoints (user management, quarter management) restricted to it_admin and corp_approver

4.2 File Upload & Smart Matching

FR-UPLOAD: File Upload

- Drag-and-drop upload zone supporting multiple files (up to 20 per upload, 50MB per file)
- Files are stored on the server filesystem (container volume); Phase 2 moves to Azure Blob Storage with SAS URLs
- Individual file upload buttons next to each required file slot for single-file targeted upload
- Uploads allowed only when submission status is draft or rejected
- File metadata recorded: original filename, file size, content type, uploaded by (user), upload timestamp

FR-MATCH: Smart File Matching

When multiple files are uploaded at once, the system attempts to auto-match each filename to a required file type using three strategies in priority order:

1. Exact short_name match: filename contains the short_name keyword (e.g., 'trial_balance' matches item #1) — confidence 90%
2. Item number match: filename contains the item number pattern (e.g., 'item_13a_rebates.xlsx') — confidence 85%

3. Fuzzy keyword match: tokenized filename compared against required_files names using Levenshtein distance — confidence varies

Matched files are presented in a confirmation modal showing the filename, matched required file, and confidence score. Unmatched files show a dropdown for manual assignment. The user reviews and confirms all mappings before files are committed.

4.3 N/A Acknowledgment Workflow

FR-NA: Not Applicable Acknowledgment

This is the core feature that solves the original business problem: explicitly recording which files are not needed for a given BU in a given quarter.

- When a BU Submitter clicks Submit to Corporate Finance with unuploaded files remaining, a modal dialog appears
- The modal lists every file still showing 'Not Uploaded' status
- Each file has a checkbox that must be checked to confirm it is not applicable
- A warning banner explains this creates an auditable record
- The Confirm button activates only when all items are checked
- Each N/A acknowledgment records: who acknowledged, when, and an optional reason
- N/A acknowledgments are stored in submission_files.is_not_applicable with na_acknowledged_by and na_acknowledged_at
- Corporate Finance can view all N/A acknowledgments in the review screen (purple highlighted section)
- N/A status can be undone (reverted to 'Not Uploaded') while the submission is still in draft

4.4 Submission Workflow

FR-SUBMIT: Submission State Machine

The system enforces strict state transitions at the submission level:

From	To	Trigger	Business Rules
draft	submitted	BU clicks Submit	All 31 files must be either uploaded or marked N/A. N/A acknowledgment form completed for all missing files.
submitted	approved	Corp Finance approves	Only corp_approver role. Submission is locked; all files transition to approved status.
submitted	rejected	Corp Finance rejects	Must provide a rejection reason. Specific files can be flagged. Rejection reason displayed to BU user.
rejected	draft	BU acknowledges rejection	Rejected files unlocked for re-upload. Previously approved files remain. BU can fix and resubmit.
submitted	draft	BU withdraws	Only allowed before Corp Finance acts. All submitted files revert to uploaded status.

FR-FILESTATUS: File-Level States

- not_uploaded: No file has been uploaded for this required item
- uploaded: File uploaded; submission is still in draft
- submitted: Submission sent to Corp Finance; file is locked from changes
- approved: Corp Finance approved this specific file
- rejected: Corp Finance rejected this file; can be re-uploaded after rejection

4.5 Corporate Finance Dashboard

FR-DASHBOARD: Submission Dashboard

- Matrix view showing all business units with their submission status for the selected quarter
- Summary statistics row: Total BUs, Approved, Pending Review, In Progress, Rejected
- Stacked completion bar showing overall progress across all BUs (green/blue/amber/red)
- Filter by segment dropdown (Electronics, Engraving, Engineering Tech, Scientific, Specialty Solutions)
- Filter by status dropdown (Draft, Submitted, Approved, Rejected)
- Quarter selector to switch between fiscal quarters
- Each row shows: BU name, code, segment, status badge, uploaded count, N/A count, progress bar, Review button
- Review button appears only for submitted BUs; opens the detailed review screen

4.6 Review & Approval

FR-REVIEW: Corp Finance Review Screen

- Header shows BU name, submitter name, submission date, and current status
- Summary cards: Files Submitted count, Marked N/A count, Completion percentage
- N/A Acknowledgments section (purple): lists all files marked N/A with who acknowledged them and when
- Rejection reason display (red): if previously rejected, shows the rejection reason and reviewer name
- Submitted Files table: item number, required file name, category, status, uploaded filename
- Decision section: Approve Submission and Reject Submission buttons
- Approve requires confirmation dialog; transitions all submitted files to approved
- Reject requires a text reason (mandatory); transitions submission and all submitted files to rejected status
- Corp Reviewer role sees Reject button only; Corp Approver sees both Approve and Reject

4.7 Email Notifications

FR-EMAIL: Automated Notifications

Event	Recipients	Content

Submission Sent	Corp Finance team	[BU Name] submitted Q[X] close binder files. [N] files uploaded, [M] marked N/A.
Submission Approved	BU submitter + team	Your Q[X] close binder submission has been approved by [Reviewer].
Submission Rejected	BU submitter + team	Your Q[X] submission was rejected. Reason: [reason]. Files flagged: [list].
Due Date Reminder	BU submitters (3 days before)	Reminder: Q[X] close binder files are due on [date]. [N] files still pending.
Overdue Alert	BU submitters + Corp Finance	[BU Name] has [N] overdue files for Q[X]. Due date was [date].
Password Reset	Requesting user	Password reset link with single-use token (1-hour expiry).

Email is sent via SMTP (nodemailer). Supports Microsoft 365, Google Workspace, SendGrid, and Amazon SES. If SMTP is not configured, the server logs [Email skipped] and continues without error.

4.8 Administration

FR-ADMIN: Administrative Functions

- Users & Assignments: Create users, assign roles, assign BUs with segment quick-select toggle, reset passwords, activate/deactivate users
- Quarter Management: Create new quarters, set quarter end dates, activate/deactivate quarters (only one active at a time)
- Due date reminders: Nightly cron job (7 AM) checks for submissions with files due within 3 days and sends reminder emails
- Audit Log: Every action (upload, submit, approve, reject, mark_na, withdraw, replace) recorded with user, timestamp, IP address, and user agent
- Health Endpoint: GET /api/health returns server status, database connection, and uptime

5. Data Model

PostgreSQL 16 database with 10 tables, indexes, and automatic updated_at triggers.

5.1 Entity Overview

Table	Purpose
business_units	77 divisions with code, name, segment, currency, parent hierarchy, active flag
users	User accounts with email, display name, role, bcrypt password hash, active flag
user_business_units	Junction table: maps each user to their permitted business units
quarters	Fiscal quarters with year, number, end date, and active flag
required_files	31 close binder file types with item number, name, short_name (for matching), category
file_due_dates	Per-file per-quarter due dates computed from quarter end + business day offset
submissions	One per BU per quarter: status (draft/submitted/approved/rejected), submitter, reviewer, rejection reason
submission_files	One per file per submission: upload status, blob path, filename, N/A flag, acknowledgment metadata
password_resets	Single-use password reset tokens with 1-hour expiry
audit_log	Every state transition: user, action, detail, IP address, user agent, timestamp

5.2 Key Relationships

- users ↔ business_units: many-to-many via user_business_units (BU access control)
- business_units → submissions: one-to-many (one submission per BU per quarter, enforced by unique constraint)
- quarters → submissions: one-to-many (submissions belong to a quarter)
- submissions → submission_files: one-to-many (31 file slots per submission)
- required_files → submission_files: one-to-many (each file slot maps to a required file type)
- required_files + quarters → file_due_dates: per-file per-quarter due dates
- All state transitions → audit_log: comprehensive action trail

5.3 Database Indexes

- idx_submissions_bu_quarter: business_unit_id + quarter_id (fast submission lookup)
- idx_submission_files_submission: submission_id (fast file listing per submission)
- idx_user_bus_user: user_id (fast BU access check)
- idx_user_bus_bu: business_unit_id (fast user listing per BU)
- idx_audit_submission: submission_id (audit trail per submission)
- Unique constraints on: business_units.code, users.email, submissions(bu_id, quarter_id), submission_files(submission_id, required_file_id)

6. API Specification

RESTful API built with Node.js 20, Express, with Zod request validation on all endpoints. All endpoints (except login and health) require JWT Bearer token authentication.

6.1 Authentication Endpoints

Method	Path	Description
POST	/api/auth/login	Authenticate with email + password; returns JWT token and user profile
GET	/api/auth/me	Get current user profile with assigned business units
PUT	/api/auth/password	Change password (requires current password)
POST	/api/auth/forgot-password	Generate password reset token; sends email or returns token if SMTP not configured
POST	/api/auth/reset-password	Reset password using single-use token

6.2 Submission & File Endpoints

Method	Path	Description
GET	/api/submissions?quarter_id=N	List submissions for user's BUs (or all for Corp Finance) in a quarter
POST	/api/submissions	Create/get a draft submission for a BU + quarter combination
GET	/api/submissions/:id/files	List all 31 files in a submission with status, metadata, due dates
POST	/api/submissions/:id/upload	Upload file(s) via multipart form; returns auto-match results
PUT	/api/submissions/:id/files/:sfld/map	Confirm file mapping (assign uploaded temp file to a required file slot)
PUT	/api/submissions/:id/files/:sfld/remove	Remove an uploaded file (revert slot to not_uploaded)
PUT	/api/submissions/:id/files/:sfld/na	Mark/unmark a required file as N/A
POST	/api/submissions/:id/submit	Submit to Corp Finance (validates all files uploaded or N/A)
POST	/api/submissions/:id/withdraw	Withdraw a submitted-but-not-reviewed submission
POST	/api/submissions/:id/approve	Corp Finance: approve the submission (corp_approver only)
POST	/api/submissions/:id/reject	Corp Finance: reject with reason (corp_approver or corp_reviewer)

6.3 Reference & Admin Endpoints

Method	Path	Description
GET	/api/quarters	List quarters (active quarter highlighted)
GET	/api/business-units	List BUs the user has access to (or all for Corp/Admin)

GET	/api/required-files	List all 31 required file types with categories
GET	/api/health	Health check: server status, DB connection, uptime
POST	/api/admin/quarters	Create a new quarter with due dates (admin only)
PUT	/api/admin/quarters/:id	Activate/deactivate a quarter (admin only)
GET	/api/admin/users	List all users with BU assignments (admin only)
POST	/api/admin/users	Create a new user with role and password (admin only)
PUT	/api/admin/users/:id/bus	Update BU assignments for a user (admin only)
PUT	/api/admin/users/:id/reset-pw	Reset a user's password (admin only)

7. Non-Functional Requirements

7.1 Security

- All passwords stored as bcrypt hashes (cost factor 10)
- JWT tokens with configurable expiry (default 8 hours)
- Helmet.js security headers on all responses
- CORS configured to allow only the application domain
- Rate limiting: 20 login attempts per 15 minutes, 500 API requests per 15 minutes
- File type validation: blocks dangerous extensions (.exe, .bat, .cmd, .sh, .ps1, .vbs, .js)
- Maximum file size: 50MB per file (configurable via MAX_FILE_SIZE_MB)
- PostgreSQL connections via SSL when deployed on Azure (auto-detected by hostname)
- Firewall (UFW): only ports 22 (SSH), 80 (HTTP redirect), 443 (HTTPS) open; PostgreSQL port 5432 blocked from external access
- SSH key-only authentication (password login disabled)
- All audit log entries include IP address and user agent for forensic review

7.2 Performance

- Target: <500ms response time for all API endpoints under normal load
- Concurrent users: designed for 10-20 simultaneous users (B2s VM or 1 vCPU Container App)
- PostgreSQL connection pool: 10 connections with auto-reconnect
- Database indexes on all foreign keys and common query patterns
- File uploads stream directly to disk (not buffered in memory)
- Frontend SPA: single-page load, then API calls for data (no full page reloads)

7.3 Reliability & Availability

- PM2 process manager auto-restarts Node.js on crash (VM deployment)
- Azure Container Apps: auto-restart, scaling 1-3 replicas, health probe on /api/health
- Graceful shutdown: application handles SIGTERM/SIGINT, closes database pool, completes in-flight requests
- Database backups: nightly pg_dump with 30-day retention (VM) or Azure-managed automatic backups (Container Apps)
- Application logs via PM2 (VM) or Azure Container Apps log streaming

7.4 Scalability

- Current capacity: 77 BUs, 31 files each, ~2,400 submission_files rows per quarter
- Database can handle 10x growth without schema changes
- File storage: local filesystem (current), Azure Blob Storage (Phase 2) for unlimited scalability
- Container Apps: horizontal scaling from 1 to 3 replicas based on CPU/memory utilization
- Azure cost remains under $150/month even at 10x current volume

7.5 Audit & Compliance

- Every state transition logged in audit_log table with: user ID, action type, detail, IP address, user agent, timestamp
- N/A acknowledgments create an auditable record: who confirmed, when, and optional reason
- Submission metadata: who submitted, when; who reviewed, when; rejection reasons preserved
- Database updated_at triggers on all tables ensure timestamps are always current
- No data is permanently deleted; deactivation flags used for users and BUs

8. Technology Stack

Layer	Technology
Frontend	Vanilla JavaScript SPA (single index.html), DM Sans + JetBrains Mono fonts, responsive CSS
Backend	Node.js 20, Express, bcryptjs, jsonwebtoken, Zod validation, Helmet, multer (file uploads)
Database	PostgreSQL 16 with pg connection pool, auto-SSL for Azure, migration scripts, seed data
Email	nodemailer with SMTP (Microsoft 365, SendGrid, Amazon SES, Google Workspace)
Container	Docker (node:20-alpine), multi-stage build, non-root user, docker-compose with PostgreSQL + Nginx
Hosting	Azure Container Apps + Azure Database for PostgreSQL Flexible Server + Azure Container Registry
Reverse Proxy	Nginx with security headers, 50MB upload limit, static asset caching
Process Mgmt	PM2 (VM deployment) or Azure Container Apps health probes and auto-restart
DevOps	Docker build + push to ACR, Azure CLI deployment, zero-downtime rolling updates

8.1 Project File Structure

File	Purpose
src/server.js	Express app with Helmet, CORS, rate limiting, graceful shutdown
src/db/pool.js	PostgreSQL connection pool with health check and auto-SSL for Azure
src/db/migrate.js	Creates all 10 tables, indexes, and updated_at triggers
src/db/seed.js	Loads 77 BUs, 31 files, 10 users with bcrypt passwords, demo data
src/middleware/auth.js	JWT generation/verification, role authorization, BU access checks
src/middleware/validate.js	Zod schemas for all endpoints (login, submissions, file mapping, admin)
src/routes/auth.js	Login, profile, password change, forgot/reset password
src/routes/submissions.js	Full workflow: list, upload, map, N/A, submit, withdraw, approve, reject
src/routes/admin.js	Quarter management, user CRUD, BU assignments, dashboard stats
src/routes/reference.js	Quarters, business units, required files, health check
src/services/email.js	SMTP email: submission/approval/rejection notices, due date reminders
src/services/audit.js	Audit log writer with IP/user agent, paginated log reader
src/utils/fileMatcher.js	Smart matching: short_name, item number, Levenshtein fuzzy, multi-file dedup
scripts/entrypoint.js	Container startup: waits for DB, runs migrations, optionally seeds
scripts/send-reminders.js	Standalone cron script for nightly due date reminder emails
public/index.html	Production SPA frontend with JWT auth, all screens, toast notifications
Dockerfile	Multi-stage Node 20 Alpine build, non-root user, health check
docker-compose.yml	App + PostgreSQL 16 + Nginx containers for local development
.env.example	Configuration template: DB, JWT, SMTP, file storage, CORS settings

9. Deployment Architecture

9.1 Production Deployment (Azure Container Apps)

Component	Configuration
Container App	standex-closebinder2, 1 vCPU / 2 GB RAM, min 1 / max 3 replicas, external ingress on port 3001
PostgreSQL	Azure Database for PostgreSQL Flexible Server, Burstable B1ms, PostgreSQL 17, West US 2
Container Registry	standexclosebinder.azurecr.io, Basic tier, admin access enabled
HTTPS	Azure-managed SSL certificate (automatic), no manual certbot needed
Environment Vars	NODE_ENV, JWT_SECRET, DB_HOST/PORT/NAME/USER/PASSWORD, SMTP_*, UPLOAD_DIR, CORS_ORIGIN
Startup	entrypoint.js: waits for DB connection, runs migrations, starts Express server
Updates	docker build + push to ACR, then az containerapp update (zero-downtime rolling update)

9.2 Alternative Deployments

- Azure VM: Ubuntu 22.04, B2s (2 vCPU, 4 GB RAM), PostgreSQL 16 local, Nginx reverse proxy, PM2, Let's Encrypt HTTPS (~$40/month)
- Docker Compose (local): Same Docker image + PostgreSQL container + Nginx container, runs on any Mac/Linux/Windows with Docker Desktop
- Mac Local: Direct Node.js + PostgreSQL via Homebrew, for development and testing

9.3 Estimated Monthly Cost

Resource	Tier	Monthly Cost
Container App	1 vCPU, 2 GB RAM	~$30
PostgreSQL Flexible	Burstable B1ms	~$15
Container Registry	Basic tier	~$5
Azure Files (uploads)	~10 GB	~$1
TOTAL		**~$51/month**

10. Future Roadmap

10.1 Phase 2 Enhancements

Feature	Description
Azure AD SSO	Replace JWT password auth with Azure AD / Entra ID single sign-on via MSAL.js (frontend) and passport-azure-ad (backend). Role claims and BU assignments map to Azure AD groups.
SharePoint Sync	Azure Function triggered on approval to copy approved files to a SharePoint folder structure organized by quarter/BU. Provides Corp Finance with familiar file browsing in SharePoint.
Azure Blob Storage	Move file uploads from container filesystem to Azure Blob Storage with SAS URL direct upload. Eliminates file size routing through the API server and provides unlimited storage.
Persistent Storage	Mount Azure Files volume to Container App for persistent upload storage across container restarts and scaling events.
Enhanced Matching	Machine learning-based file matching using historical upload patterns per BU. Learns from confirmed mappings to improve accuracy over time.
Audit Trail Viewer	Full-featured audit log UI in the Admin section with filtering by user, action, date range, and BU. Export to CSV for compliance review.
Due Date Scheduler	Azure Container Apps scheduled job (replacing cron) for automated due date reminder emails at scale.
Reporting Dashboard	Historical analytics: quarter-over-quarter completion rates, average submission times, overdue trends by segment.

10.2 Known Limitations (Current Release)

- File uploads are stored on the container filesystem; files are lost if the container is replaced without persistent volume
- Email notifications require SMTP configuration; without it, notifications are silently skipped
- Due date reminders require an external cron scheduler (not yet configured on Azure Container Apps)
- No file preview or in-browser document viewing; files must be downloaded to review
- Single active quarter at a time; bulk operations across quarters not supported
- No CSV/Excel export of dashboard data (planned for Phase 2)

Appendix A: Business Unit Segments

Segment	BU Count	Description
Electronics	27	Reed switches, relays, sensors, magnetics — global operations across US, Mexico, Canada, UK, Germany, China, Japan, India
Engraving	31	Mold texturizing, tool finishing, surface engineering — Americas, EMEA, and Asia Pacific
Engineering Technologies	6	Metal spinning, deep drawing, custom hoists — Spincraft, McStarlite, Custom Hoists
Scientific	2	Scientific refrigeration and storage — Horizon Scientific, Custom Biogenic Systems
Specialty Solutions	1	Refrigerated merchandising — Federal Industries
Corporate Services	10	Headquarters, shared services, corporate entities
TOTAL	**77**	